EXTERN + OUTAMATION
EXTERNSHIP

ADVANCED AI-POWERED

DOCUMENT INSIGHT AND DATA EXTRACTION PROJECT

SPRINT 6 STATUS UPDATE

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SKILLS

- Python
- AI/ML
- Data Scrapping
- Prompt Engineering
- LLMs
- Text Extraction
- Document Intelligence
- Python Automation
- Optical Character Recognition (OCR)
- Retrieval-Augumented Generation (RAG)

Helping businesses improve, automate, and architect critical tasks (such as document handling, customer service, and data management) by saving thousands of hours and boosting accuracy.

Mortgage Documents

SPRINTS

2025

01 AI Document Intelligence
02 Python & Google Colab
03 Python Data Extraction
04 Optimizing OCR
05 Implementing RAG
06 Optimizing RAG Pipelines
07 Blob Processing & Classification
08 Interactive Chatbot
09 AI-Powered Document Automation Platform

PROJECT TIMELINE

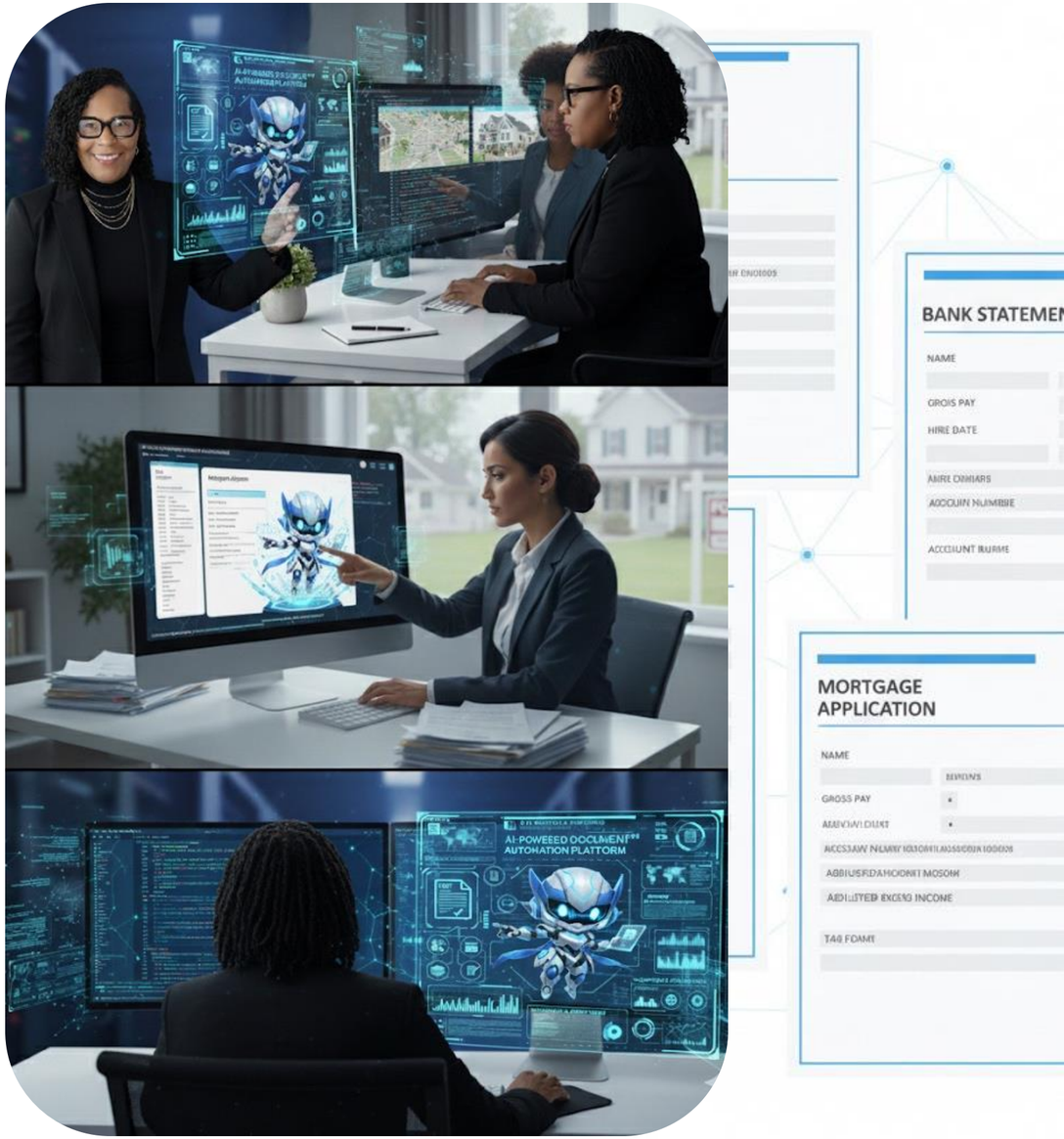
10 WEEKS

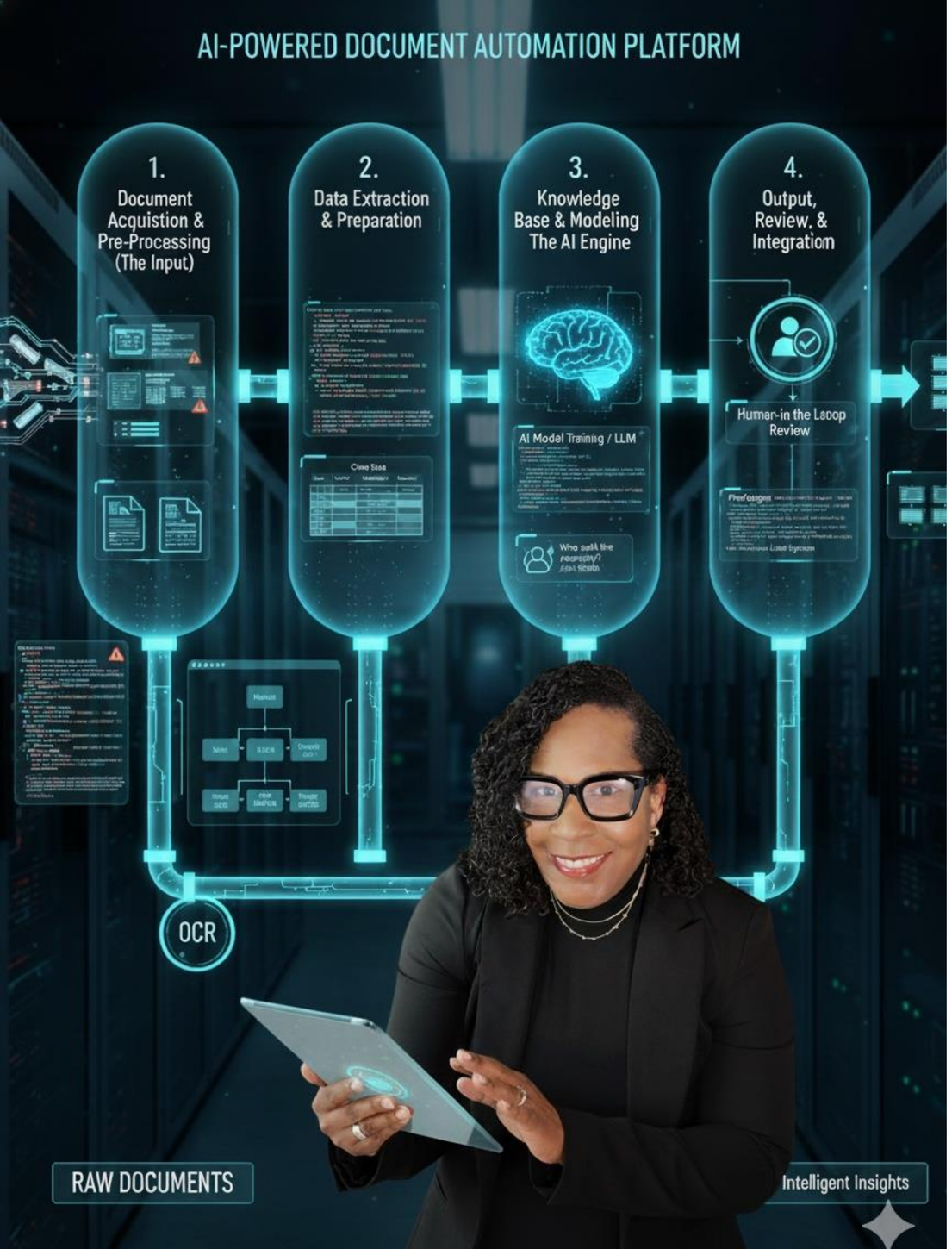
Optimizing RAG Pipelines

Open-source LLMs: Mistral and Phi-2

Background: 10-Week Project

The Outamation Advanced AI-Powered Document Insights and Data Extraction Externship is a comprehensive 10-week program. Its core goal and purpose are to master the required skills to create an AI-Powered Document Automation Platform. The program focuses on bridging foundational technologies (ML, LLMs, NLP, Computer Vision) with high-value industry applications, such as solving complex, document-heavy challenges in mortgage automation, ultimately transforming slow, manual review processes into automated, efficient workflows.





Challenge & Solution: Building The Document Intelligence (AI) Pipeline

- **The Business Challenge:**
  - Automating the highly manual and time-consuming process of mortgage underwriting by handling 200-300 pages of unstructured documents.

- **The Solution:**
  - Designing and implementing a robust, AI-Powered Document Automation Platform to transform raw files into structured, intelligent insights.

- **The 4-Step Document Intelligence Workflow:**
  - Step 1. Document Acquisition & Pre-Processing (The Input)
  - Step 2. Data Extraction & Preparation
  - Step 3. Knowledge Base & Modeling (The AI Engine)
  - Step 4. Output, Review, & Integration (The Results)

SPRINT 6 FOCUS

Step 4. Output, Review, & Integration (The Results)

Sprint 6 will concentrate on finalizing the user-facing solution and integrating the AI-powered insights back into the core underwriting process. Sprint 6 shifts the focus from the backend AI architecture to the front-end user experience and critical data integration, ensuring the intelligent insights from the AI Engine (Sprint 5 focus) deliver immediate business value

Overall Status: Sprint 6 Complete

Sprint 6 focused on taking a baseline Retrieval-Augmented Generation (RAG) pipeline to the next level by significantly improving retrieval precision and evaluating the efficacy of open-source Large Language Models (LLMs).

The key outcomes demonstrated a crucial trade-off between the speed and retrieval quality of embedding models, and confirmed that strategic retriever tuning (specifically using a reranker) and the deployment of efficient open-source models (like Mistral-7B in GGUF format) can build a high-performing, cost-effective, and private document analysis system.

Open-Source Embedding Comparison

The reported performance metrics for the embedding models represent a single instance of execution, as results can exhibit variation dependent on the time of deployment and environmental factors.

Model	Average Indexing Time (s)	Average Retrieval Time (s)	Context Conciseness Score	Key Feature / Trade-off
BGE-small-en	1.26s (Fastest)	0.10s (Fast)	**5/5 (Cleanest Context)**	**Balanced Winner:** Quickest setup and superior semantic focus, minimizing irrelevant 'noise'.
E5-small-v2	2.35s (Slowest)	**0.08s (Fastest)**	4/5 (Pulled Noise)	**Query Speed Champion:** Ideal for high-volume, real-time lookups where setup is infrequent.
MiniLM-L6-v2	1.63s (Moderate)	0.10s (Fast)	4/5 (Pulled Noise)	Competitive speed but compromised retrieval precision, consistently pulling in "noise" (e.g., unrelated refund sections).

Embedding Analysis

- The BGE-small-en model was selected as the balanced winner.
- Although E5-small-v2 offered the fastest query time, BGE-small-en's consistent 5/5.
- Conciseness Score made it the preferred choice for applications prioritizing high-quality, clean results by exhibiting superior semantic focus and minimizing the risk of feeding irrelevant information to the LLM.

Fine-Tuning Retriever Parameters

Optimizing the retriever mechanism ensures that the best possible context is selected before it is passed to the LLM.

The project experimented with three configurations:

1. Varying top_k (retrieval recall).
2. Varying the similarity threshold (precision filtering).
3. Introducing a local reranker (second-stage quality check).

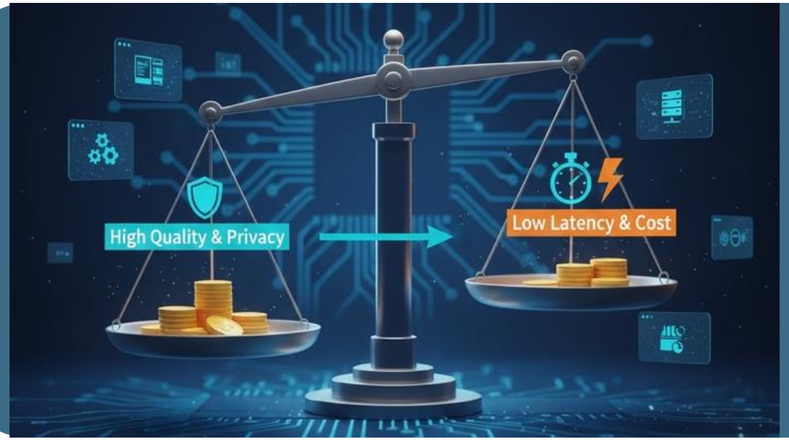
The results showed that Experiment D (Rerank) yielded the best result for quality and precision. Integrating the BGE-reranker-base cross-encoder provided a superior second-stage filter, aggressively keeping only the best nodes and significantly minimizing irrelevant context for the LLM.

Evaluating Open-Source LLMs in the RAG Pipeline

For specialized tools, open-source AI models offer distinct advantages over proprietary options, including data privacy(essential for legal/financial documents), cost savings (no per-request fees), and the ability to customize the AI.

Key open-source models evaluated for RAG include:

- **Mistral-7B (GGUF):** Considered the efficient choice, optimized for RAG pipelines and Q&A, offering a strong balance between performance and efficiency. The GGUF format is crucial for efficient, optimized LLM inference on consumer hardware (e.g., Colab GPUs) by quantizing and compressing the model.
- **Phi-2:** A lightweight and fast option, best for quick experiments and testing RAG pipelines with minimal setup



Evaluation Trade-off

- **Quality vs. Footprint:** Achieving the highest generation quality (e.g., against proprietary models like Gemini) requires using a larger computational footprint, which inherently results in slightly longer latency than the fastest models tested.

- **Strategic Trade-off:** This latency increase is considered a necessary and acceptable trade-off to gain critical enterprise advantages: full control over the environment, guaranteed data privacy, and a significantly lower, predictable cost-r-inference provided by the open-source pipeline.

CoLab Notebook

- RAG with Open-Source Model: Mistral 7B
  - https://github.com/LashawnFofung/RAG-Pipelines/blob/main/src/Task_RAG_with_Open_Source_Model_Mistral_7B.ipynb

- LLM Evaluation for RAG:
  - Hhttps://github.com/LashawnFofung/RAG-Pipelines/blob/main/src/Task_LLM_Evaluation_for_RAG.ipynb

THANK YOU

https://github.com/LashawnFofung/AI-Powered-Document-Automation-Platform